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| <b>Course Title</b>       | <b>Web Technologies</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| <b>Course Code</b>        | <b>DI-322</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Credit Hours</b>       | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| <b>Category</b>           | Domain Core                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Prerequisite</b>       | CC-112: Programming Fundamentals                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| <b>Co-Requisite</b>       | None                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Follow Up</b>          | Enterprise Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| <b>Course Description</b> | <b>Introduction:</b> Web Applications, TCP/IP Application Services. <b>Web Servers:</b> Basic Operation, Virtual hosting, Chunked transfers, Caching support, Extensibility. SGML, HTML5, CSS3. <b>XML Languages and Applications:</b> Core XML, XHTML, XHTML MP. <b>Web Services:</b> SOAP, REST, WML, XSL. Operations, Processing HTTP Requests, Processing HTTP Responses, Cookie Coordination, Privacy and P3P, Complex HTTP Interactions, Dynamic Content Delivery. Server Configuration. Server Security. <b>Web Browsers Architecture and Processes:</b> Active Browser Pages: JavaScript, DHTML, AJAX. JSON. <b>Approaches to Web Application Development:</b> Programing in any Scripting language. Search Technologies, Search Engine Optimization. XML Query Language, Semantic Web, Future Web Application Framework. |
| <b>Text Book(s)</b>       | 1. Paul J. Deitel and Harvey Deitel, Java How to Program, 11 <sup>th</sup> Edition, Pearson, 2017, ISBN-10: 0134743350, ISBN-13: 978-0134743356.<br>2. Marty Hall and Larry Brown, Core Servlets and JavaServer Pages, 2 <sup>nd</sup> Edition, Pearson, 2017, ISBN-10: 8131701638, ISBN-13: 978-8131701638.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| <b>Reference Material</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

